

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
END TERM EXAMINATION-NOVEMBER 2024

Course: B.Tech

Subject Name: NETWORK ANALYSIS

Time Duration: 3 Hours

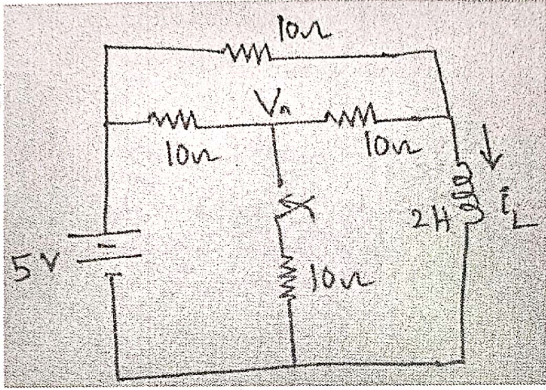
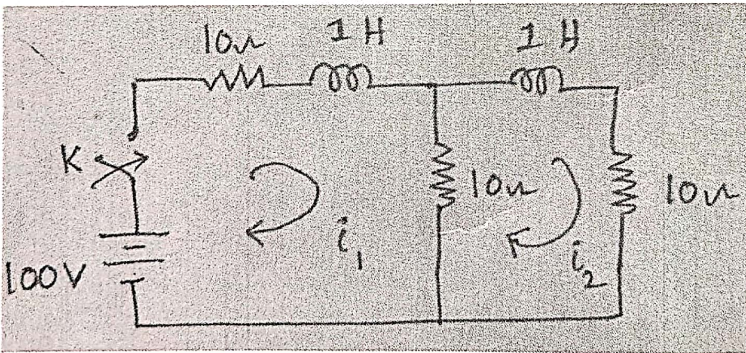
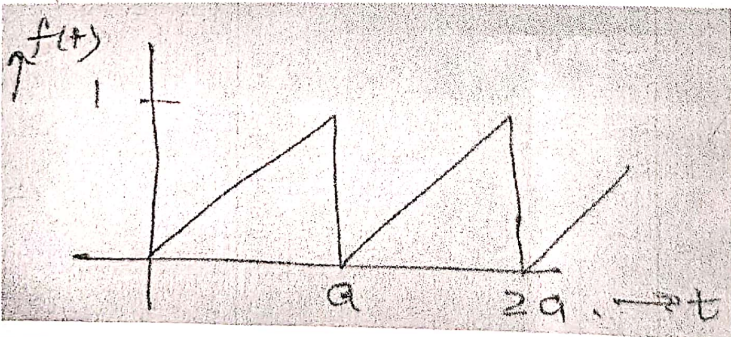
Subject Coordinator Name: Dr. Savita Nema and Dr. More Raju

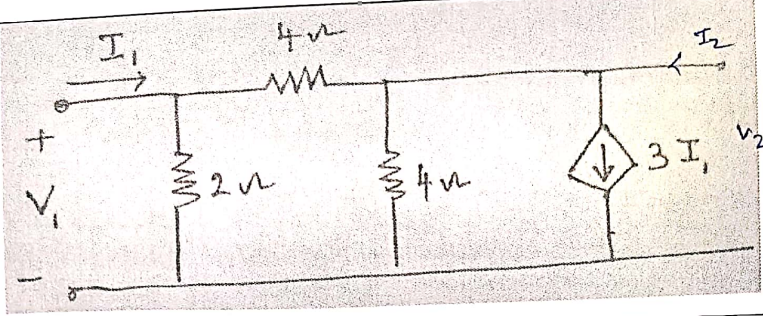
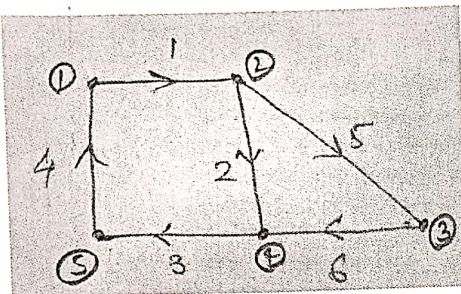
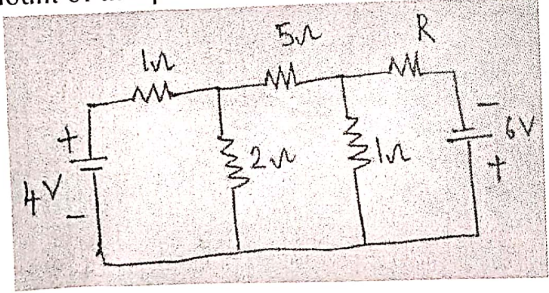
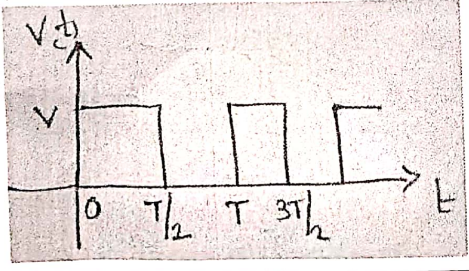
Note: Assume necessary data if required.

Semester: III

Subject Code: EE-213

Maximum Marks: 50

S.No	Question	Marks	CO
1	For the circuit shown, switch is closed at t. Find $V_a(0^-)$ and $V_a(0^+)$. 	5	CO-2
2	In the following circuit switch is closed at $t = 0$. Find $i_2(s)$ and $i_2(t)$ using Laplace transformation. 	5	CO-2
3	Find the Laplace transform of the waveform shown below. 	5	CO-2
4	Find Y and Z parameters of the network shown.	5	CO-3

			
5	Write fundamental cutset matrix for the network shown. Tree branches are 1, 2, 3, 6. Write KCL and KVL equations. 	5	CO-1
6	Synthesize the following RC impedance function using Foster and Cauer forms. $F(s) = \frac{3(s+2)(s+4)}{s(s+3)}$	10	CO-5
7	A) Define maximum power transfer theorem for a purely resistive circuit with variable load and prove it. B) Find the value of R in the following circuit such that maximum power transfer takes place. Also find the amount of this power. 	5	CO-1
8	Find the Fourier series expansion for the periodic waveform shown below. 	5	CO-4